UCT732: Advanced Social, Text and Media Analytics

L	T	P	Cr
2	**0**	**2**	**3.0**

Course Objectives: To provide an understanding of various text mining operations, preprocessing techniques, content analysis, sentiment analysis, web analytics tools and social media analytics.

Text Mining: Introduction, Core text mining operations, Pre-processing techniques, Categorization, Clustering, Information extraction, Probabilistic models for information extraction, Text mining applications

Methods & Approaches: Content Analysis; Natural Language Processing; Clustering & Topic Detection; Simple Predictive Modelling; Sentiment Analysis; Sentiment Prediction

Web Analytics: Web analytics tools, Clickstream analysis, A/B testing, online surveys; Web search and retrieval, Search engine optimization, Web crawling and Indexing, Ranking algorithms, Web traffic models

Social Media Analytics: Social network and web data and methods. Graphs and Matrices. Basic measures for individuals and networks. Information visualization; Making connections: Link analysis. Random graphs and network evolution. Social contexts: Affiliation and identity; Social network analysis

Laboratory Work:
Usage of R or Python to carry out text mining operations, sentiment analysis along with sentiment prediction. Insight to Web Analytics by the use of Google Analytics and Adobe Analytics along with Social Media Analytics tools.

Course Learning Outcomes (CLOs) / Course Objectives (COs):
Upon completion of this course, students will be able to:

1. Understand the concept of Text Mining along its operations and its usages in different applications
2. Apply and interpret methods for performing Content Analysis, Clustering, Predictive modeling and Sentiment Analysis.
3. Illustrate the usage of various Web Analytics Tool for carrying out analysis, testing and optimization of Websites.
4. Comprehend the basis of Web Traffic Models, Ranking Algorithms, Information Visualization and Social Media Analytics on various social media sites.

Text Books:

1. Ronen Feldman and James Sanger, “The Text Mining Handbook: Advanced Approaches in Analyzing Unstructured Data”, Cambridge University Press, 2006.

2. Hansen, Derek, Ben Sheiderman, Marc Smith. 2011 Analyzing Social Media Networks with NodeXL: Insights from a Connected World, Morgan Kaufmann, 304.
3. Avinash Kaushik. 2009. Web Analytics 2.0: The Art of Online Accountability.
4. Hanneman, Robert and Mark Riddle. 2005. Introduction to Social Network Method.

Reference Books:

1. Wasserman, S. & Faust, K. (1994). Social network analysis: Methods and applications. New York: Cambridge University Press.
2. Monge, P. R. & Contractor, N. S. (2003). Theories of communication networks. New York: Oxford University Press. http://nosh.northwestern.edu/vita.html.